

Netflix Database Simulation: SQL & Relational Design

1. Introduction & System Overview

This project models a relational database simulating the core operations of the "Netflix" streaming platform, structured in Third Normal Form (3NF) to ensure data integrity and eliminate redundancies. The schema bridges subscriber demographics and onboarding interest themes with an extensive content catalog (covering formats, genres, production hubs, and original titles) alongside granular watch history and rating metrics.

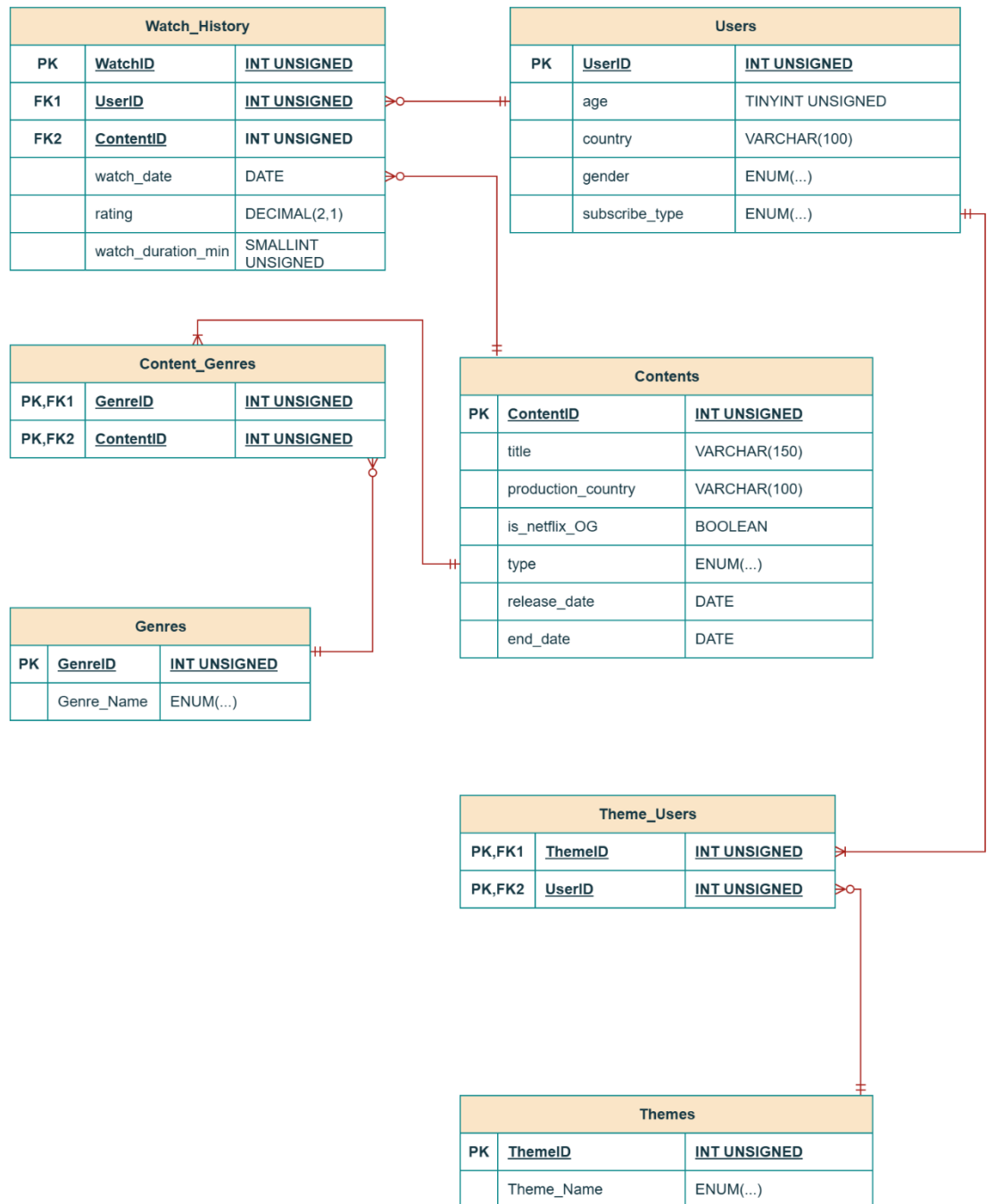
The primary objective of the database is to provide business intelligence that drives content acquisition, catalog auditing, and personalized subscriber retention. To support these strategic decisions, the database is queried across five analytical dimensions:

- **Behavioral Alignment:** Evaluating discrepancies between user-declared onboarding themes and actual genre consumption.
- **Demographic Audience Profiling:** Mapping subscriber age cohorts and satisfaction levels across specific interest categories.
- **Production Hub Efficiency:** Benchmarking normalized viewing performance and critical ratings of Netflix Originals across international production markets.
- **Episodic Catalog Performance:** Assessing viewership volume, catalog saturation, and reception across serialized formats.
- **Catalog Utilization & Dormancy Auditing:** Detecting unstreamed "dormant" titles alongside active assets to inform licensing renewals, recommendation prioritization, and content retirement strategies.

2. Working Assumptions for Diagram & Database Design:

- A registered user might not have watched any content yet.
- A user can watch the same content item multiple times (e.g., a user watching "Shrek 2" 5 times).
- Multiple users can watch the same content item.
- A content item can exist in the catalog without ever being watched.
- A genre can exist before any corresponding content is added to the system (e.g., the genre "Sci-Fi" exists without an uploaded movie).
- Every content item must have at least one genre.
- A single content item can belong to multiple genres (e.g., a series categorized as both "Drama" and "Based on a True Story").
- Every user must select at least one theme.
- A theme can exist without being selected by any user.
- The same theme can be selected by multiple users.

3. Database Architecture & ERD



4. Normalized Database Schema (3NF)

- **1NF (First Normal Form):** Every column across all tables stores strictly atomic, single-valued entries (numbers, text, dates) without delimited lists. Each table defines a unique single or composite Primary Key (PK).
- **2NF (Second Normal Form):**
 - The schema satisfies 1NF.
 - In single-PK tables (Users, Contents, Genres, Themes, Watch_History), all non-key attributes are fully functionally dependent on the primary key.
 - In composite-PK bridge tables (Content_Genres and Theme_Users), all columns form the composite key itself, eliminating partial key dependencies.
- **3NF (Third Normal Form):**
 - The schema satisfies 2NF.
 - No transitive dependencies exist between non-key attributes. User attributes (age, country, gender) depend strictly on UserID.
 - Genres and themes are decoupled into dedicated lookup tables (Genres, Themes) connected via bridge tables, avoiding redundancy.

5. Research Questions & Analytics (Queries 1–5)

Query 1: Cross-Sectional Alignment Between User Declared Themes and Actual Watched Genres

Business Motivation: Evaluates the correlation between user declared themes upon registration and actual viewing habits across genres (view count, total watch time, and average session duration). This enables Netflix to detect behavioral discrepancies, such as lifestyle theme subscribers consuming high intensity genres, and optimize recommendation algorithms accordingly.

SQL Query:

```
SELECT
    t.theme_name,
    g.Genre_Name,
    COUNT(wh.watchID) AS total_views,
    SUM(wh.watch_duration_min) AS total_watch_minutes,
    ROUND(AVG(wh.watch_duration_min), 1) AS avg_duration_per_view
FROM themes t
JOIN theme_users tu ON t.themeID = tu.themeID
JOIN watch_history wh ON tu.userID = wh.userID
JOIN content_genre cg ON wh.contentID = cg.contentID
JOIN genres g ON cg.genreID = g.genreID
GROUP BY t.theme_name, g.Genre_Name
ORDER BY t.theme_name ASC, total_watch_minutes DESC;
```

Table 1: Query 1 Output - User Declared Themes vs. Watched Genres Breakdown

theme_name	Genre_Name	total_views	total_watch_minutes	avg_duration_per_view
Cooking	Sci-Fi	2	100	50.0
Fashion	Thriller	1	154	154.0
Fashion	Sci-Fi	1	55	55.0
Religion	Drama	1	58	58.0
Science	Action	1	132	132.0
Science	Documentary	1	50	50.0
Sports	Drama	1	148	148.0
Sports	Documentary	1	50	50.0
History	Drama	2	206	103.0
Music	Thriller	1	154	154.0
Music	Action	1	132	132.0
Music	Sci-Fi	1	45	45.0

Analytical Insights:

- **Preference vs. Consumption Divergence:** The data reveals a clear discrepancy between stated preferences and actual viewing, as subscribers declaring lifestyle themes (such as Cooking, Fashion, and Music) heavily consume high-intensity genres (such as Thriller, Action, and Sci-Fi).
- **History Retention Lead:** History enthusiasts demonstrate the highest overall engagement and session retention, accumulating 206 minutes across Drama titles with an exceptional average of 103.0 minutes per view.

Query 2: Demographic Profiling and Viewer Engagement Across Themes

Business Motivation: Analyzes subscriber age distributions, viewing volume, and content satisfaction ratings across declared interest themes. This segmentation enables Netflix to profile the target audience for each category, identify engagement hotspots versus content dissatisfaction risks, and strategically allocate acquisition and production budgets based on demographic preferences.

SQL Query:

```
SELECT
    t.theme_name,
    COUNT(DISTINCT tu.userID) AS interested_users,
    ROUND(AVG(u.age), 1) AS avg_age,
    COUNT(wh.watchID) AS total_watches,
    ROUND(AVG(wh.rating), 2) AS avg_rating
FROM themes t
JOIN theme_users tu ON t.themeID = tu.themeID
JOIN users u ON tu.userID = u.userID
LEFT JOIN watch_history wh ON u.userID = wh.userID
GROUP BY t.theme_name
ORDER BY total_watches DESC;
```

Table 2: Query 2 Output - Subscriber Demographics & Satisfaction Metrics per Theme

theme_name	interested_users	avg_age	total_watches	avg_rating
Music	3	23.0	3	3.90
Cooking	2	26.5	2	4.50
Fashion	2	23.5	2	4.25
Science	2	25.0	2	4.35
Sports	2	32.5	2	4.95
History	2	39.5	2	4.60
Religion	1	45.0	1	4.20

Analytical Insights:

- **Demographic Cohort Segmentation:** Younger subscribers dominate lifestyle categories such as Music (mean age 23.0) and Fashion (23.5), whereas older demographics concentrate in History (39.5) and Religion (45.0).
- **Music Catalog Gap:** While Music drove the highest user interest and watch volume, it registered the lowest satisfaction score (3.90 rating), signaling a potential quality gap or catalog mismatch for younger demographics.
- **Mature Cohort Resonance:** Mature audiences exhibit superior satisfaction and content resonance, led by Sports (4.95 rating) and History (4.60 rating), underscoring strong retention efficiency within these dedicated segments.

Query 3: Normalized Performance of Netflix Originals by Production Country

Business Motivation: Evaluates viewer engagement and satisfaction for Netflix Original productions across global production hubs. By analyzing content performance relative to catalog size (`views_per_title`), this query moves beyond raw volume to provide a normalized, cross-sectional assessment. Applying a quality threshold (`avg_rating >= 4.0`) isolates international production markets that consistently deliver high-satisfaction original assets.

SQL Query:

```
SELECT
    c.production_country,
    COUNT(DISTINCT c.contentID) AS total_originals,
    COUNT(wh.watchID) AS total_watch,
    ROUND(COUNT(wh.watchID) / COUNT(DISTINCT c.contentID), 1) AS
views_per_title,
    ROUND(AVG(wh.rating), 2) AS avg_rating
FROM contents c
JOIN watch_history wh ON c.contentID = wh.contentID
WHERE c.is_netflix_OG = 1
GROUP BY c.production_country
HAVING avg_rating >= 4.0
ORDER BY views_per_title DESC;
```

Table 3: Query 3 Output - Normalized Originals Performance by Production Country

production_country	total_originals	total_watch	views_per_title	avg_rating
United Kingdom	3	3	1.0	4.55
United States	1	1	1.0	4.50

Analytical Insights:

- **Normalized Viewing Efficiency:** When normalizing views relative to catalog size (`views_per_title`), both the United Kingdom and the United States achieve an identical ratio of 1.0 view per title, confirming balanced audience adoption across both production markets.
- **UK Volume and Rating Lead:** The United Kingdom serves as the primary original production hub (3 titles) and edges ahead in critical reception with an exceptional 4.55 average rating (compared to 4.50 for the US), demonstrating a proven capacity to deliver multi-title volume while sustaining premium viewer satisfaction.

Query 4: Viewer Engagement and Catalog Performance Across Series Genres

Business Motivation: Evaluates subscriber engagement and satisfaction specifically for television series across distinct catalog genres. By contrasting catalog availability (total_series) with actual viewing activity (total_watch and avg_rating), this query isolates high-affinity narrative genres from lower-performing episodic categories. These insights guide series renewal decisions, episodic catalog expansion, and production budget allocations toward genres demonstrating superior viewer retention and critical reception.

SQL Query:

```
SELECT
    g.Genre_Name,
    COUNT(DISTINCT c.contentID) AS total_series,
    COUNT(wh.watchID) AS total_watch,
    ROUND(AVG(wh.rating), 2) AS avg_rating
FROM genres g
JOIN content_genre cg ON g.genreID = cg.genreID
JOIN contents c ON cg.contentID = c.contentID
JOIN watch_history wh ON c.contentID = wh.contentID
WHERE c.`type` = 'series'
GROUP BY g.Genre_Name
HAVING total_watch >= 1
ORDER BY total_watch DESC, avg_rating DESC;
```

Table 4: Query 4 Output - Episodic Catalog Engagement and Ratings by Genre

Genre_Name	total_series	total_watch	avg_rating
Sci-Fi	2	2	4.50
Documentary	1	1	4.90
Drama	1	1	4.20

Analytical Insights:

- **Sci-Fi Volume Driver:** Sci-Fi serves as the primary episodic anchor in the active catalog, housing 2 titles and generating the highest viewing engagement (2 watches) while sustaining a strong 4.50 satisfaction score.
- **Documentary Quality Benchmark:** While Documentary accounts for a single active title, it attained the highest critical reception across all television genres with a 4.90 rating, indicating strong audience appreciation and retention potential for specialized factual programming.

- **High Overall Satisfaction:** All active episodic genres maintain elevated satisfaction thresholds (≥ 4.20), confirming consistent production quality across the serialized portfolio. Future catalog investments should prioritize expanding high-satisfaction niches like Documentary alongside proven anchors like Sci-Fi.

Query 5: Catalog Utilization Audit and Dormancy Identification

Business Motivation: Evaluates catalog asset activation and identifies dormant content items that generate zero subscriber engagement. While active titles receive sustained viewer traction, unviewed assets tie up catalog space without providing viewing value. By auditing all catalog titles against watch history records, this analysis isolates unstreamed content alongside performing titles to assess catalog health and guide promotional or licensing retention decisions.

SQL Query:

```
SELECT
    c.title,
    g.Genre_Name,
    c.`type`,
    COUNT(wh.watchID) AS total_watch
FROM genres g
JOIN content_genre cg ON g.genreID = cg.genreID
JOIN contents c ON cg.contentID = c.contentID
LEFT JOIN watch_history wh ON c.contentID = wh.contentID
GROUP BY c.contentID, c.title, g.Genre_Name, c.`type`
ORDER BY total_watch ASC, c.title ASC;
```

Table 5: Query 5 Output - Catalog Utilization and Dormancy Breakdown

title	Genre_Name	type	total_watch
The Silent Valley	Sci-Fi	movie	0
Black Mirror	Sci-Fi	series	1
Inception	Drama	movie	1
Our Planet	Documentary	series	1
Parasite	Action	movie	1
Pulp Fiction	Thriller	movie	1
Stranger Things	Sci-Fi	series	1
The Crown	Drama	series	1

Analytical Insights:

- **Dormancy Isolation:** The audit identifies *The Silent Valley* (Sci-Fi, Movie) as the catalog's sole dormant asset with 0 total views, establishing an overall portfolio dormancy rate of 12.5% (1 out of 8 titles).
- **Balanced Catalog Activation:** The remaining 87.5% of the catalog demonstrates consistent baseline engagement (1 view per title), distributed evenly across episodic and feature formats (4 series and 3 active movies).
- **Sci-Fi Category Contrast:** Sci-Fi contains both top-tier serialized anchors (*Stranger Things* and *Black Mirror*) and the only unviewed asset (*The Silent Valley*), indicating that dormancy is title-specific rather than a reflection of genre disinterest.

6. Project Deliverables / Files

- `Netflix_ERD` - Entity-Relationship Diagram (ERD).
- `netflix_project_db.sql`- SQL schema DDL for the Netflix database.
- `netflix_project_data.sql`- SQL DML data insertion script.
- `netflix_project_queries.sql`- SQL research analytical queries.